L8: Critical Density Threshold in CFinder

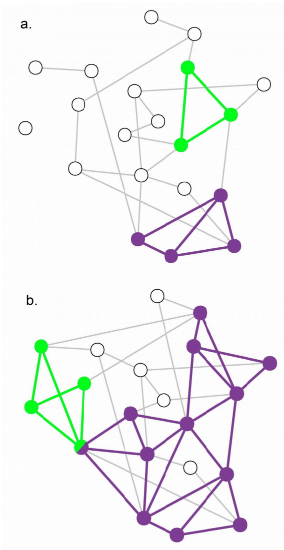

Figure 9.22 from Network Science by Albert-László Barabási

An interesting question about the CFinder algorithm is whether it would detect communities even in completely random networks. A completely random network should not have any community structure.

Suppose that k=3. If the network is sufficiently dense, it will have many triangles formed strictly based on chance, and so the algorithm would detect a number of communities even though the network structure is completely random.

So we can ask the following question: *for a given network size n and value of k, what is the maximum density of a random network of size n so that the appearance of k-cliques is unlikely?* Let's call this density as the "critical density threshold".

Obviously, the higher k is, the higher the critical density threshold should be.

It is not hard to prove that the critical density threshold is:

$$p_c(k) = [(k-1)n]^{\frac{-1}{k-1}}$$

When the density is lower than $p_c(k)$we expect only few isolated k-cliques. If the network density exceeds $p_c(k)$, we observe numerous cliques that form k-clique communities.

For k=2 the k-cliques are simply individual links and the critical density threshold reduces to $p_c(k) = 1/n$, which is the condition for the emergence of a giant connected component in Erdős–Rényi networks.

For k =3 the k-cliques become triangles and the critical density threshold is $p_c(k) = 1/\sqrt{2n}$

The visualization shows two random networks: one with density p=0.13 (part-a) and the other with density p=0.22 (part-b). For this network size (n=20) and for k=3, the critical density threshold is 0.16. This means that the network of part-a is below the threshold, and indeed there are only three triangles in that network. Cfinder would report in that case that most network nodes do not belong to any community.

The network of part-b on the other hand is above the critical density threshold and it includes many triangles. Cfinder would report that all but four nodes of that network belong to communities. From the statistical perspective, this would be an incorrect conclusion.

The moral of the story is that before we apply the Cfinder algorithm, we should also check whether the density of the network is below or above the critical threshold for the selected value of k. If it is larger than the threshold, we should consider larger values of k (even though this would make the algorithm more computationally intensive).

Food for Thought

Derive the previous expression for the critical density threshold.

Hint: To have a giant connected component of k-cliques, the average k-clique should have at least two adjacent k-cliques (this is known as Molloy-Reed criterion -- further discussed in Lesson-11). Any lower network density than the critical threshold will result in a lower average than two adjacent k-cliques to a randomly chosen k-clique.

So, consider a given k-clique X and suppose it has an adjacent k-clique. At the critical density value the expected number of additional adjacent k-cliques to X should be one. So, start by deriving the expression for the expected number of additional adjacent k-cliques to X, as a function of k and p. Then set that expectation equal to 1, and solve for p to derive the critical density threshold. You can simplify the expression assuming that n is very large.